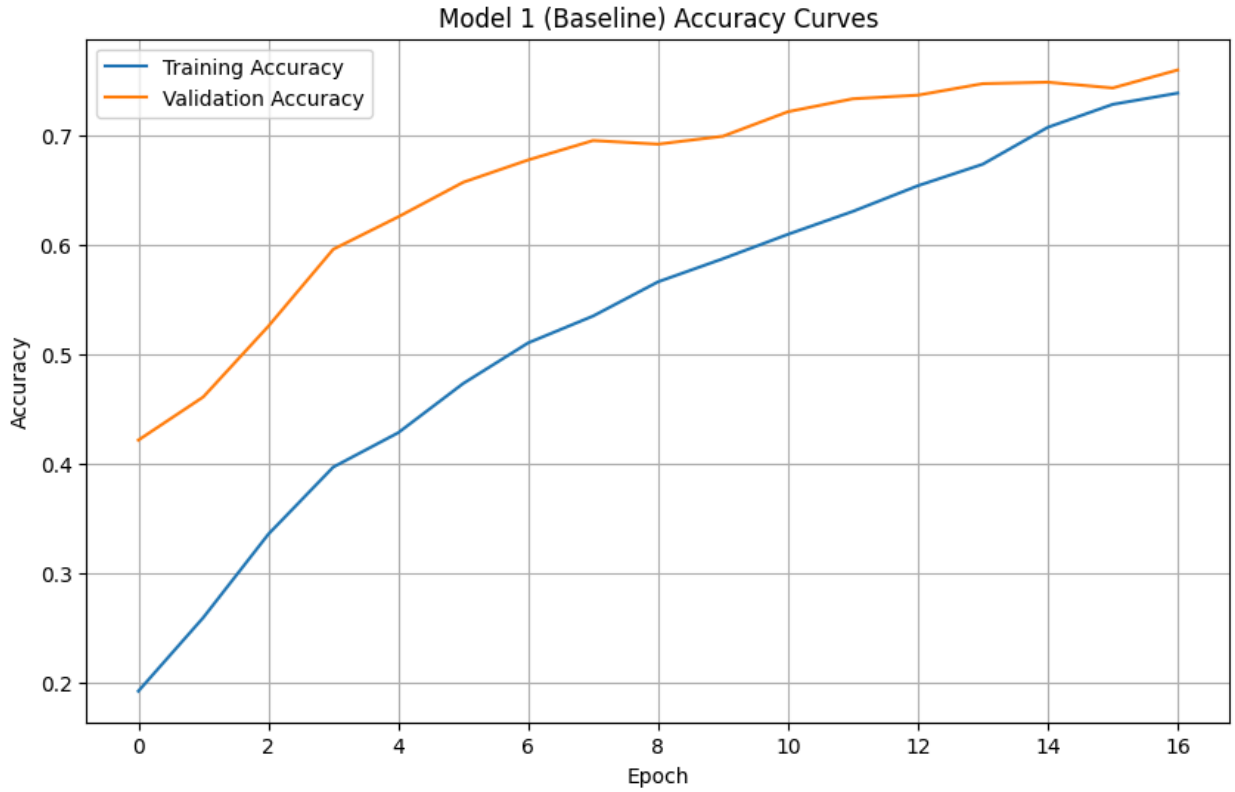
Model 1: EfficientNetB0 (BASELINE)

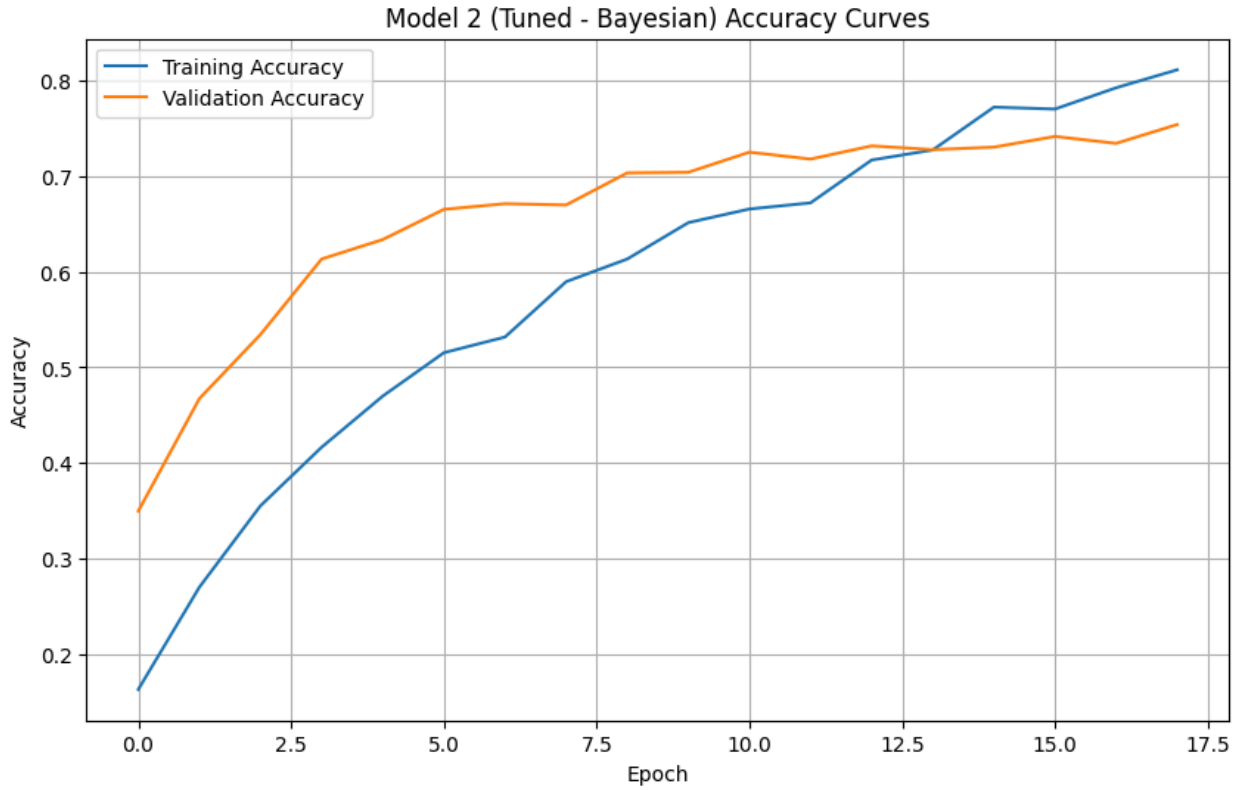



```
WARNING:tensorflow:Model failed to serialize as JSON. Ignoring... Unable to
serialize [2.0896919 2.1128857 2.1081853] to JSON. Unrecognized type <class
'tensorflow.python.framework.ops.EagerTensor'>.
Epoch 1/20
219/219 [==============================] - 67s 222ms/step - loss: 2.4326 - accuracy:
0.1925 - val_loss: 1.5691 - val_accuracy: 0.4220
Epoch 2/20
219/219 [==============================] - 49s 217ms/step - loss: 1.7420 - accuracy:
0.2598 - val_loss: 1.4803 - val_accuracy: 0.4615
Epoch 3/20
219/219 [==============================] - 39s 174ms/step - loss: 1.3903 - accuracy:
0.3359 - val_loss: 1.3304 - val_accuracy: 0.5260
Epoch 4/20
219/219 [==============================] - 37s 163ms/step - loss: 1.1662 - accuracy:
0.3973 - val_loss: 1.1611 - val_accuracy: 0.5964
Epoch 5/20
219/219 [==============================] - 35s 159ms/step - loss: 1.0063 - accuracy:
0.4287 - val_loss: 1.0641 - val_accuracy: 0.6261
Epoch 6/20
219/219 [==============================] - 36s 158ms/step - loss: 0.8861 - accuracy:
0.4737 - val_loss: 0.9610 - val_accuracy: 0.6577
Epoch 7/20
219/219 [==============================] - 36s 159ms/step - loss: 0.7408 - accuracy:
0.5109 - val_loss: 0.9377 - val_accuracy: 0.6781
Epoch 8/20
219/219 [==============================] - 36s 159ms/step - loss: 0.7143 - accuracy:
0.5354 - val_loss: 0.9067 - val_accuracy: 0.6959
Epoch 9/20
219/219 [==============================] - 36s 159ms/step - loss: 0.6141 - accuracy:
```

```
0.5667 - val_loss: 0.8880 - val_accuracy: 0.6926
Epoch 10/20
219/219 [==============================] - 36s 159ms/step - loss: 0.5426 - accuracy:
0.5878 - val_loss: 0.8631 - val_accuracy: 0.6998
Epoch 11/20
219/219 [==============================] - 36s 159ms/step - loss: 0.5342 - accuracy:
0.6101 - val_loss: 0.8226 - val_accuracy: 0.7222
Epoch 12/20
219/219 [==============================] - 36s 159ms/step - loss: 0.4957 - accuracy:
0.6311 - val_loss: 0.7878 - val_accuracy: 0.7340
...
Epoch 16/20
219/219 [==============================] - 36s 160ms/step - loss: 0.3286 - accuracy:
0.7289 - val_loss: 0.7554 - val_accuracy: 0.7439
Epoch 17/20
219/219 [==============================] - 36s 160ms/step - loss: 0.3102 - accuracy:
0.7392 - val_loss: 0.7284 - val_accuracy: 0.7604
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Model 2: EfficientNetB0 (Tuned)




```
Reloading Tuner from logs\skin_tuning_bayesian\tuner0.json
Retraining model dengan hyperparameter terbaik...
Epoch 1/20
219/219 [==============================] - 48s 182ms/step - loss: 2.4472 - accuracy:
0.1628 - val_loss: 1.6483 - val_accuracy: 0.3496
Epoch 2/20
219/219 [==============================] - 38s 170ms/step - loss: 1.7056 - accuracy:
0.2699 - val_loss: 1.4650 - val_accuracy: 0.4674
Epoch 3/20
219/219 [==============================] - 38s 170ms/step - loss: 1.2765 - accuracy:
0.3552 - val_loss: 1.2342 - val_accuracy: 0.5346
Epoch 4/20
219/219 [==============================] - 40s 173ms/step - loss: 1.0486 - accuracy:
0.4165 - val_loss: 1.0692 - val_accuracy: 0.6136
Epoch 5/20
219/219 [==============================] - 37s 168ms/step - loss: 0.9169 - accuracy:
0.4700 - val_loss: 1.0169 - val_accuracy: 0.6340
Epoch 6/20
219/219 [==============================] - 37s 165ms/step - loss: 0.7910 - accuracy:
0.5154 - val_loss: 0.9498 - val_accuracy: 0.6656
Epoch 7/20
219/219 [==============================] - 37s 165ms/step - loss: 0.7172 - accuracy:
0.5318 - val_loss: 0.9039 - val_accuracy: 0.6715
Epoch 8/20
219/219 [==============================] - 37s 164ms/step - loss: 0.5954 - accuracy:
0.5898 - val_loss: 0.9276 - val_accuracy: 0.6702
Epoch 9/20
219/219 [==============================] - 37s 164ms/step - loss: 0.5572 - accuracy:
0.6135 - val_loss: 0.8199 - val_accuracy: 0.7038
```

```
Epoch 10/20
219/219 [==============================] - 37s 164ms/step - loss: 0.4709 - accuracy:
0.6517 - val_loss: 0.8353 - val_accuracy: 0.7044
Epoch 11/20
219/219 [==============================] - 37s 164ms/step - loss: 0.4349 - accuracy:
0.6660 - val_loss: 0.7939 - val_accuracy: 0.7255
Epoch 12/20
...
Epoch 17/20
219/219 [==============================] - 37s 164ms/step - loss: 0.2536 - accuracy:
0.7929 - val_loss: 0.8530 - val_accuracy: 0.7347
Epoch 18/20
219/219 [==============================] - 37s 165ms/step - loss: 0.2294 - accuracy:
0.8118 - val_loss: 0.8375 - val_accuracy: 0.7544
```

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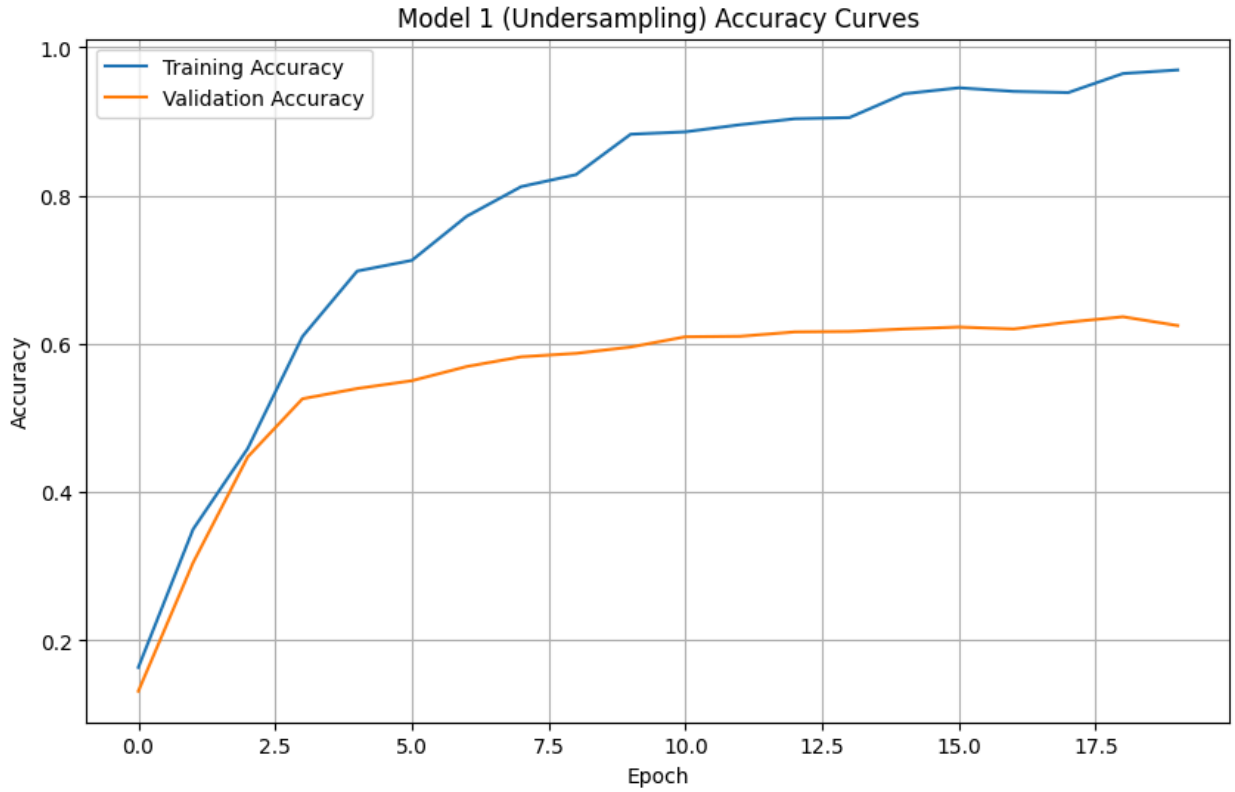
Model 1: Undersampling Models (BASELINE)

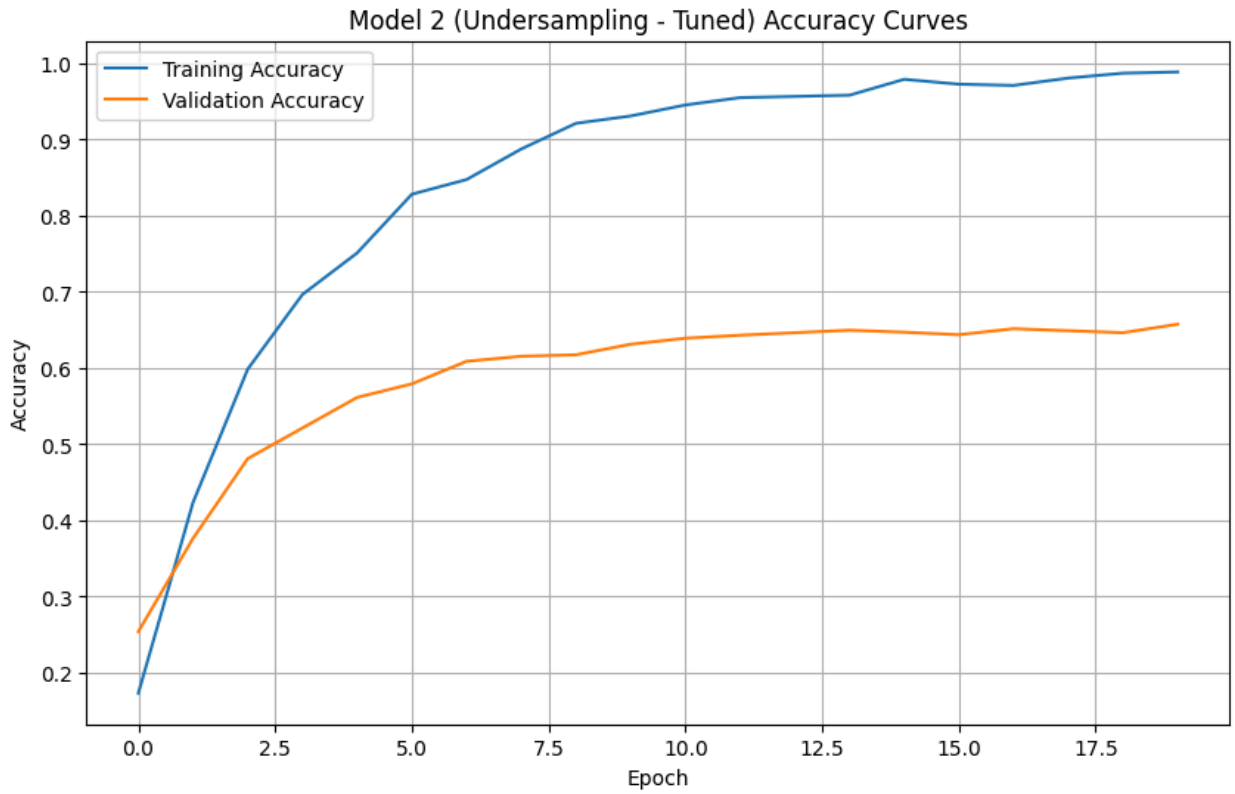



```
Epoch 1/20
20/20 [==============================] - 15s 382ms/step - loss: 2.8891 - accuracy:
0.1637 - val_loss: 1.9462 - val_accuracy: 0.1317
Epoch 2/20
20/20 [==============================] - 6s 291ms/step - loss: 2.0010 - accuracy:
0.3499 - val_loss: 1.7769 - val_accuracy: 0.3048
Epoch 3/20
20/20 [==============================] - 6s 292ms/step - loss: 1.4743 - accuracy:
0.4591 - val_loss: 1.6071 - val_accuracy: 0.4477
Epoch 4/20
20/20 [==============================] - 6s 293ms/step - loss: 1.1380 - accuracy:
0.6100 - val_loss: 1.4571 - val_accuracy: 0.5260
Epoch 5/20
20/20 [==============================] - 6s 293ms/step - loss: 0.9048 - accuracy:
0.6982 - val_loss: 1.3581 - val_accuracy: 0.5398
Epoch 6/20
20/20 [==============================] - 6s 293ms/step - loss: 0.7831 - accuracy:
0.7127 - val_loss: 1.2872 - val_accuracy: 0.5504
Epoch 7/20
20/20 [==============================] - 6s 293ms/step - loss: 0.6650 - accuracy:
0.7721 - val_loss: 1.2260 - val_accuracy: 0.5695
Epoch 8/20
20/20 [==============================] - 6s 293ms/step - loss: 0.5516 - accuracy:
0.8122 - val_loss: 1.1647 - val_accuracy: 0.5826
Epoch 9/20
20/20 [==============================] - 6s 294ms/step - loss: 0.4962 - accuracy:
0.8283 - val_loss: 1.1196 - val_accuracy: 0.5872
Epoch 10/20
20/20 [==============================] - 6s 295ms/step - loss: 0.3789 - accuracy:
```

```
0.8828 - val_loss: 1.1077 - val_accuracy: 0.5958
Epoch 11/20
20/20 [==============================] - 6s 297ms/step - loss: 0.3644 - accuracy:
0.8860 - val_loss: 1.0750 - val_accuracy: 0.6096
Epoch 12/20
20/20 [==============================] - 6s 295ms/step - loss: 0.3447 - accuracy:
0.8957 - val_loss: 1.0668 - val_accuracy: 0.6103
Epoch 13/20
...
Epoch 19/20
20/20 [==============================] - 6s 293ms/step - loss: 0.1477 - accuracy:
0.9647 - val_loss: 1.1005 - val_accuracy: 0.6366
Epoch 20/20
20/20 [==============================] - 6s 293ms/step - loss: 0.1340 - accuracy:
0.9695 - val_loss: 1.1375 - val_accuracy: 0.6248
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Model 2: Undersampling Models (Tuned)




```
Epoch 1/20
20/20 [==============================] - 14s 362ms/step - loss: 2.5899 - accuracy:
0.1734 - val_loss: 1.8186 - val_accuracy: 0.2541
Epoch 2/20
20/20 [==============================] - 6s 302ms/step - loss: 1.6084 - accuracy:
0.4238 - val_loss: 1.6933 - val_accuracy: 0.3766
Epoch 3/20
20/20 [==============================] - 6s 303ms/step - loss: 1.1355 - accuracy:
0.5987 - val_loss: 1.5464 - val_accuracy: 0.4812
Epoch 4/20
20/20 [==============================] - 6s 303ms/step - loss: 0.8766 - accuracy:
0.6966 - val_loss: 1.4173 - val_accuracy: 0.5214
Epoch 5/20
20/20 [==============================] - 6s 304ms/step - loss: 0.7033 - accuracy:
0.7512 - val_loss: 1.3275 - val_accuracy: 0.5616
Epoch 6/20
20/20 [==============================] - 6s 305ms/step - loss: 0.5482 - accuracy:
0.8283 - val_loss: 1.2457 - val_accuracy: 0.5793
Epoch 7/20
20/20 [==============================] - 6s 305ms/step - loss: 0.4643 - accuracy:
0.8475 - val_loss: 1.1746 - val_accuracy: 0.6090
Epoch 8/20
20/20 [==============================] - 6s 305ms/step - loss: 0.3595 - accuracy:
0.8876 - val_loss: 1.1127 - val_accuracy: 0.6155
Epoch 9/20
20/20 [==============================] - 6s 304ms/step - loss: 0.2983 - accuracy:
0.9213 - val_loss: 1.0662 - val_accuracy: 0.6175
Epoch 10/20
20/20 [==============================] - 6s 305ms/step - loss: 0.2731 - accuracy:
```

```
0.9310 - val_loss: 1.0236 - val_accuracy: 0.6313
Epoch 11/20
20/20 [==============================] - 6s 305ms/step - loss: 0.2241 - accuracy:
0.9454 - val_loss: 1.0004 - val_accuracy: 0.6392
Epoch 12/20
20/20 [==============================] - 6s 306ms/step - loss: 0.2063 - accuracy:
0.9551 - val_loss: 1.0016 - val_accuracy: 0.6432
Epoch 13/20
...
Epoch 19/20
20/20 [==============================] - 6s 305ms/step - loss: 0.0930 - accuracy:
0.9872 - val_loss: 1.0411 - val_accuracy: 0.6465
Epoch 20/20
20/20 [==============================] - 6s 306ms/step - loss: 0.0798 - accuracy:
0.9888 - val_loss: 1.0426 - val_accuracy: 0.6577
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...